The Value of a College Education

Problem Background

For this project, suppose you work as a data analyst for Glassdoor and you are tasked with writing an article that highlights college graduate salaries 5 years after graduation. For this article you obtain data from the American Community Survey Public Use Microdata Series. The dataset Salary.csv contains the following variables:

Variable Name	Description
Salary	Graduate's annual salary 5 years after graduation
MajorCategory	Graduate's primary major listed on diploma
Sex	Graduate's sex
GPA	Graduate's overall grade point average

In this project, you are asked to write an article that can be *read by a general audience* who use Glassdoor and a supporting documentation article on your analysis that can be used so other data analysts can reproduce or build upon your work. Your general audience article should be about a 5-10 min read.

Your *general audience* article should be split into the following sections and cover the listed topics. You can (and should) use professional looking figures and relevant numbers (also give intervals if appropriate for the topic) from your corresponding analysis in the article but should NOT use statistical jargon. Here is an example article that could give you a flavor of the style you are shooting for.

1. Introduction
    - Use a catchy title that can be used to get people to click on the article
    - Basic introduction and overview of what the article is about
2. Impact of GPA
    - What impact, if any, does GPA have on salary? Do you think students should be hyper-anxious about grades? Is the effect of GPA different for different majors (does GPA matter more for some majors than others)?
3. Salaries Across Majors
    - Do you see differences in salaries across majors? If so, what majors seem to make more money?
4. Salaries for Women and Men
    - Are there differences in salaries for men vs. women within the same major category? If so, does the difference in salary between men vs. women differ by major category? If there are differences, demonstrate what the difference is between men and women salary for your major category as an example.
5. Other Factors

    - How well do the variables you have in your dataset determine salary? Do you think that there are other variables that contribute to salary? Use a specific example to demonstrate how well (or not well) the factors can be used to determine (i.e. predict) salary.
6. Conclusions
    - What should be the take away message readers get from your article?

Your *documentation* article (because its for internal use only) is more informal in tone and can use statistical jargon. For the documentation, you are talking to fellow data analysts. Your documentation should have the following structure:

1. Modeling
    - Write out the math behind what model you are using to write your article. For example, $\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$) where, for reproducibility, make sure you state what is in your $\mathbf{X}$ matrix. This model should be used to back up statements in all the sections of your article.
    - Validate your model (including predictions if you are using predictions in your article - hint, you should be)
2. For each of section 2-5 above, write out:
    - Description of how you used the model to write the corresponding section of article (e.g. we looked at β_{GPA} and a CI OR we did an F-test etc.).
3. Your well-documented code

Even though the general article and documentation article are separate, for grading reasons, please just submit 1 PDF but make a clear distinction between the two articles (please clear page between the two).

Suggestions on Use of AI

You are encouraged to use AI as a collaborative partner on this project—especially for planning your analysis, understanding regression results, and writing in two different styles (general-audience article vs. technical documentation). AI should support your reasoning and communication, but it should not replace your modeling work or generate results you do not understand.

To start, you can upload the data and this assignment. To help guide AI more effectively, you might also consider uploading relevant lecture notes and/or your HW code. In general, the more background and guidance you provide, the more useful AI's responses will be.

The following prompts may help:

1. **Use AI to plan your exploration and visuals**
    - *Prompt idea:* "Based on the outline for the general-audience article, what analyses or visualizations would be appropriate, and what should I pay attention to as I explore the data?"
2. **Use AI as a conceptual tutor for regression-based questions**
    - *Prompt idea:* "Help me understand how I would use regression to answer the following question (insert question from above here). Ask me guiding questions on how to structure my analysis."
3. **Use AI for coding support, but stay in control**
    - *Prompt idea:* "I want to use (insert method here) to answer the question (insert question here). Help me generate some code in R to do this, but explain what each piece of the code does and

why it is needed."

4. **Use AI as an editor (not a results-generator)**
    - *Prompt idea:* "Help me organize, expand, and polish these sections based on the results I have already computed. Do not invent numbers or conclusions—use only the results shown in my output."
5. **Use AI to check clarity and identify gaps**
    - *Prompt idea:* "Is there anything I left out that would strengthen this analysis or make the article clearer? Please give suggestions, but do not invent results I haven't computed."
6. **Use AI to help you switch tone between the two required documents**
    - *Prompt idea:* "Help me write this in a casual, fun tone that anyone can enjoy reading. Puns are encouraged."
    - *Prompt idea:* "This section is for a technical audience. Please write it so that someone could reproduce my analysis."

Important guardrails:

- AI should help you think, question, and communicate—not do the analysis for you.
- You should be able to explain every figure, model choice, and conclusion in your own words.
- Do not include AI-generated numbers or claims unless they come directly from your own code and output.
- If AI suggests a function or method you don't recognize, ask follow-up questions and connect it back to course tools.
- Keep brief notes on how AI influenced your thinking, decisions, or writing during the project.